

Marketing A/B Testing Analysis: The Effect of Advertisement on Customer Conversion Rate

William Chen

Abstract

This report presents an A/B testing analysis of a digital marketing campaign, examining whether exposure to a commercial advertisement (*ad*) increases conversion rate compared to a public service announcement (*psa*) across 588,101 users. Both Frequentist and Bayesian approaches are applied to a binary conversion outcome (*converted*) and subgroup analyses by day and hour of peak ad exposure.

Frequentist results show a statistically significant difference in conversion rate ($\chi^2(1) = 54.006$, $p < 0.001$), with the *ad* group achieving 2.55% compared to 1.79% for *psa*. Bayesian posterior analysis confirms a 100% probability that *ad* yields a higher conversion rate, with an estimated advantage of 0.59%–0.94%. Subgroup analysis identifies Monday and the afternoon hours (14:00–20:00) as the highest-performing conditions. All effect sizes remain negligible, though the consistency across both methods supports the effectiveness of commercial advertisement exposure.

1 Introduction

Understanding the effectiveness of digital advertisement is critical for optimizing marketing strategies and maximizing return on investment. While traditional metrics such as click-through rates provide surface-level insights, conversion rate, the proportion of users who complete a purchase after exposure, serves as a more direct measure of advertisement effectiveness.

This report presents an A/B testing analysis of a digital marketing campaign, examining whether exposure to a commercial advertisement (*ad*) leads to a higher conversion rate compared to a public service announcement (*psa*). The dataset contains 588,101 user records, with users randomly assigned to one of two experimental groups.

Both Frequentist and Bayesian approaches are applied to evaluate the primary binary outcome (*converted*) and to examine conversion patterns across days and hours of peak ad exposure. The Frequentist approach determines whether observed differences are statistically significant, while the Bayesian approach provides posterior probabilities and credible intervals for a more interpretable conclusion.

2 Data

2.1 Dataset Overview

The dataset contains 588,101 user records sourced from Kaggle, collected from a randomized A/B experiment conducted on a digital marketing campaign. Users were randomly assigned to one of two experimental groups: *psa* (control, $n = 23,524$) and *ad* (treatment, $n = 564,577$). No missing values were detected across all variables. The primary outcome variable is *converted*, a binary indicator representing whether a user made a purchase after being exposed to the advertisement.

2.2 Feature Description

The dataset contains 6 variables: 1 identifier (*user.id*, excluded from analysis), 1 grouping variable (*test.group*), 1 behavioral metric (*total.ads*), 1 binary outcome variable (*converted*), and 2 temporal variables (*most.ads.day*, *most.ads.hour*). *test.group* indicates the experimen-

tal group assigned to each user, either *psa* (control) or *ad* (treatment). *total.ads* records the total number of advertisements viewed by each user. *converted* is a logical variable indicating whether a user completed a purchase. *most.ads.day* and *most.ads.hour* record the day of the week and hour of the day during which the user was most frequently exposed to advertisements, respectively.

3 Exploratory Data Analysis

3.1 Summary Statistics

The dataset contains 588,101 complete observations with no missing values detected. The two experimental groups are imbalanced by design: *psa* contains 23,524 users (4.0%) and *ad* contains 564,577 users (96.0%). The overall conversion rate is 2.52%, with *ad* achieving 2.55% and *psa* achieving 1.79%. *total.ads* ranges from 1 to 2,065 with a mean of 24.82 and a median of 13, indicating a heavily right-skewed distribution.

3.2 Conversion Rate by Test Group

Figure 1 shows the conversion rates for both experimental groups. The *ad* group achieves a higher conversion rate (2.55%) compared to the *psa* group (1.79%), indicating a descriptive difference of 0.76 percentage points. Whether this difference is statistically significant is examined in the Results section.

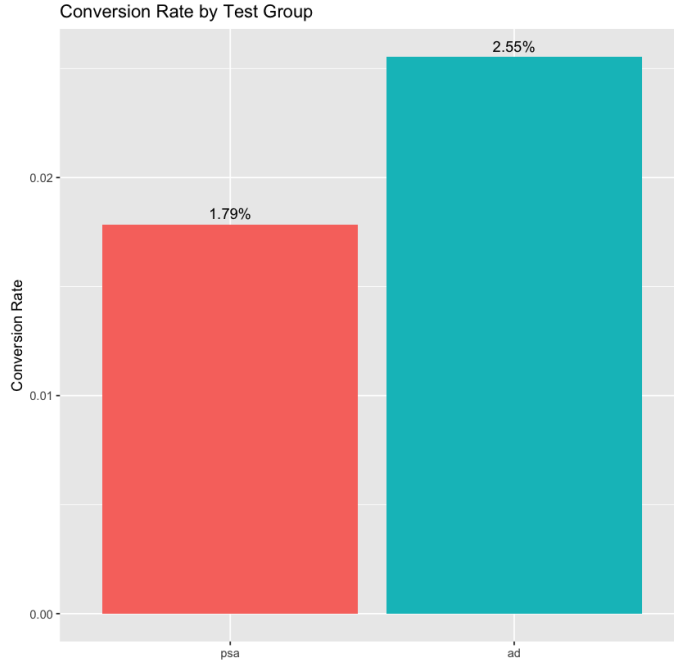


Figure 1: Conversion Rate by Test Group

3.3 Conversion Rate by Day and Hour

As shown in Figure 2, Monday consistently shows the highest conversion rate in the *ad* group, while Thursday, Friday, and Saturday perform lower. Figure 3 shows that the afternoon period (14:00–20:00) achieves the highest conversion rates, with hour 16 reaching the peak.

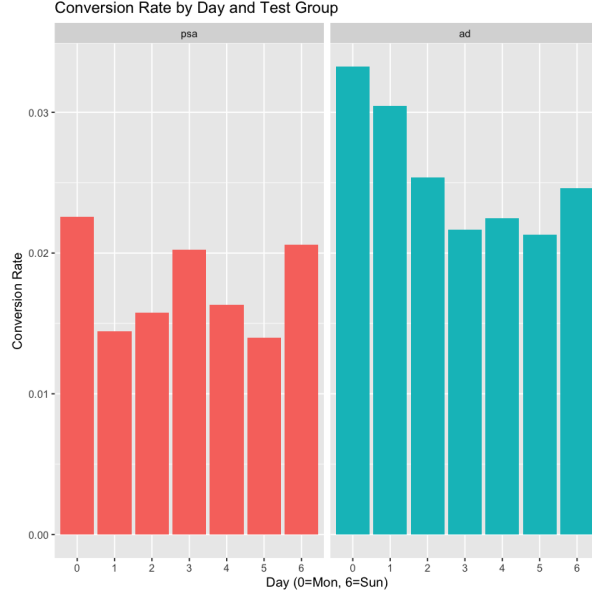


Figure 2: Conversion Rate by Day

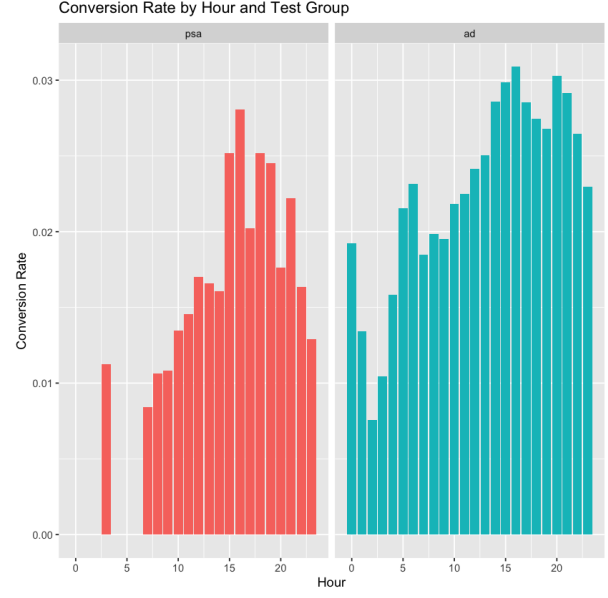


Figure 3: Conversion Rate by Hour

Figure 4 and Figure 5 present the conversion rate and sample size heatmaps for the *ad* group. The notably high conversion rate observed on Saturday during early morning hours corresponds to cells with substantially fewer observations. In low-sample cells, a small number of conversions can produce a large rate spike, making such estimates unreliable. Elevated rates in these periods are therefore more likely to reflect random fluctuation than genuine user behavior.

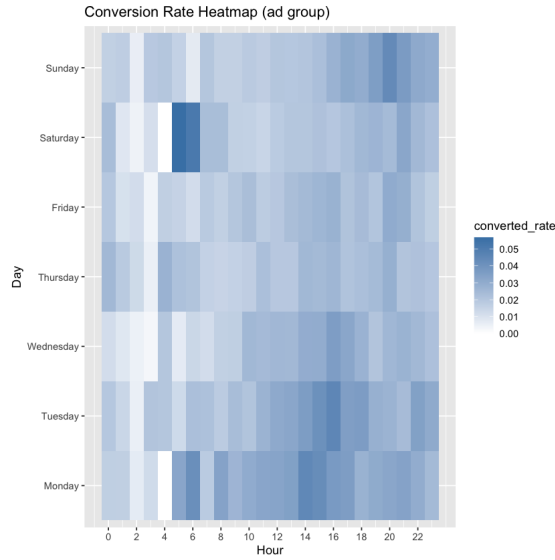


Figure 4: Conversion Rate Heatmap (ad group)

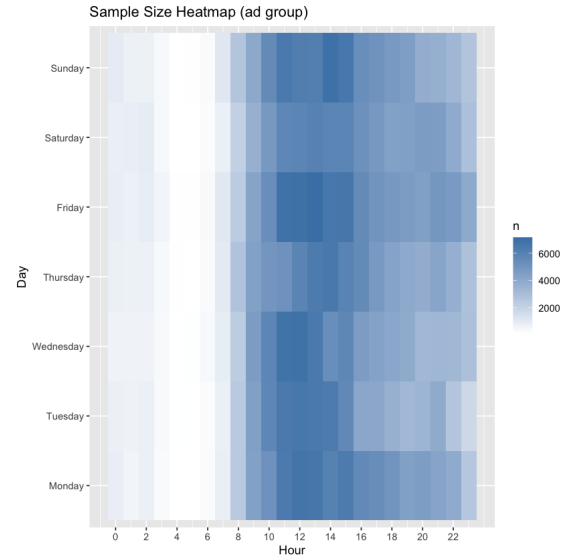


Figure 5: Sample Size Heatmap (ad group)

3.4 Outlier Detection

To assess the presence of extreme values in *total.ads*, the IQR method was applied with a threshold of $\text{IQR} \times 3$. The upper bound was determined to be 96, with 24,825 observations (4.22%) exceeding this threshold. Figure 6 visualizes the outlier threshold across both groups on a log scale.

Users exceeding the upper bound exhibit substantially higher conversion rates (17.18% for *ad*, 11.51% for *psa*) compared to non-outlier users (1.91% and 1.31%, respectively), suggesting that high ad exposure is associated with increased conversion likelihood. As these represent genuine high-engagement users rather than data errors, outliers are retained in the analysis.

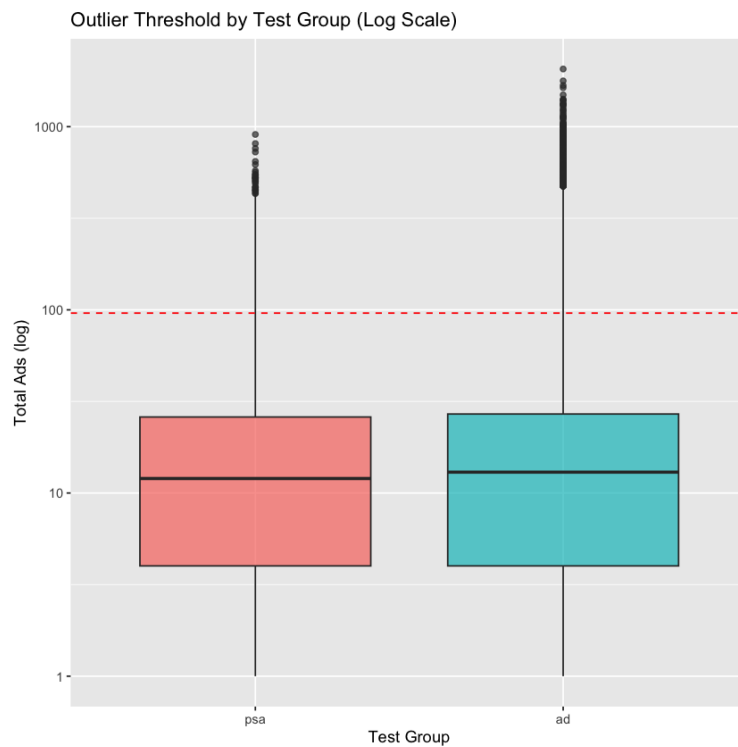


Figure 6: Total Ads Distribution by Group (Log Scale). The dashed red line indicates the $\text{IQR} \times 3$ outlier threshold.

4 Methodology

4.1 Frequentist Approach

Mann-Whitney U Test The Mann-Whitney U test was applied to compare *total.ads* between the two groups. As a non-parametric test, it does not assume normality and is therefore appropriate for the right-skewed distribution of ad exposure counts.

Two-Sample Proportion Test The two-sample proportion test was applied to evaluate whether advertisement exposure is associated with a higher conversion rate. The null hypothesis states that the conversion rate is independent of group assignment.

Chi-Square Test The Chi-square test of independence was applied to evaluate whether conversion rate varies across days and hours within the *ad* group.

4.2 Bayesian Approach

Beta-Binomial Model For binary conversion outcomes, a Beta-Binomial conjugate model was applied with an uninformative prior $\text{Beta}(1, 1)$, yielding the posterior:

$$p \mid \text{data} \sim \text{Beta}(\alpha + \text{successes}, \beta + \text{failures})$$

Gamma-Poisson Model For *total.ads*, a Gamma-Poisson conjugate model was applied with prior $\text{Gamma}(1, 1)$, yielding the posterior:

$$\lambda \mid \text{data} \sim \text{Gamma}\left(\alpha + \sum x_i, \beta + n\right)$$

4.3 Evaluation Metrics

Frequentist Frequentist tests are evaluated using p-value and effect size: Cohen’s *h* for the two-sample proportion test, Cramér’s *V* for Chi-square tests, and rank-biserial *r* for the Mann-Whitney U test.

Bayesian For Bayesian models, Monte Carlo sampling ($n = 100,000$) was used to estimate the following metrics:

- $P(ad > psa)$: posterior probability that *ad* outperforms *psa*
- **95% HDI**: the Highest Density Interval representing the most credible range of the difference between groups

5 Frequentist Results

5.1 Primary Analysis (converted)

A randomization check confirms that the two groups received comparable ad exposure (mean: 24.76 vs. 24.82), with no meaningful difference detected by the Mann-Whitney U test ($r = 0.0086$, small), supporting the validity of the experimental design.

The two-sample proportion test reveals a statistically significant difference in conversion rate between the *ad* and *psa* groups ($\chi^2(1) = 54.006$, $p < 0.001$). The *ad* group achieves a conversion rate of 2.55%, compared to 1.79% for *psa*, yielding a difference of 0.76 percentage points (95% CI: $[-0.0095, -0.0059]$). However, Cohen’s $h = 0.053$ indicates a negligible practical effect size.

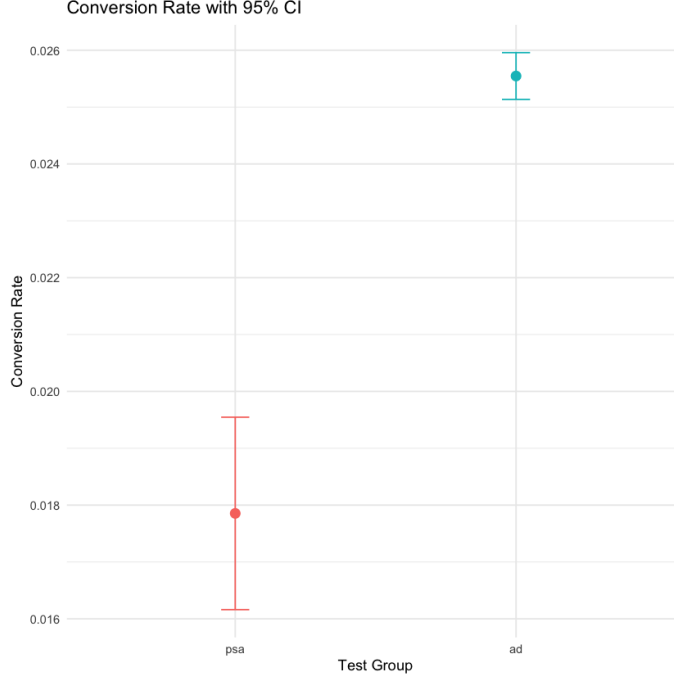


Figure 7: Conversion Rate with 95% CI by Test Group

5.2 Subgroup Analysis (ad group only)

To further examine the conditions under which advertisement exposure is most effective, conversion rates within the *ad* group are analyzed by day and hour of peak ad exposure. Table 1 presents the conversion rates by day. Monday achieves the highest conversion rate (3.32%) with the largest positive residual (+13.92), while Saturday records the lowest (2.13%, residual = -7.45).

The Chi-square test confirms a significant difference across days ($\chi^2(6) = 412.79$, $p < 0.001$, Cramér's $V = 0.027$), and pairwise tests with Bonferroni correction show that Monday differs significantly from all other days ($p \leq 0.034$), while Thursday, Friday, and Saturday show no significant differences among themselves ($p = 1.000$). Although statistically significant, Cramér's $V = 0.027$ indicates a negligible practical effect, suggesting that the difference in conversion rates across days is minimal in absolute terms (ranging from 2.13% to 3.32%).

Conversion Rate by Day (ad group)			
Day	n	Conversion Rate	Std. Residual (TRUE)
Monday	83,571	3.32%	+13.92
Tuesday	74,572	3.04%	+8.36
Wednesday	77,418	2.54%	−0.33
Thursday	79,077	2.16%	−6.88
Friday	88,805	2.25%	−5.75
Saturday	78,802	2.13%	−7.45
Sunday	82,332	2.46%	−1.66

Table 1: Conversion Rate and Standardized Residuals by Day (ad group)

Table 2 presents the conversion rates by hour within the *ad* group, focusing on the top 10 performing hours (13:00–22:00). Hour 16 achieves the highest conversion rate (3.09%, residual = +6.34), while hour 13 records the lowest (2.51%, residual = −0.64), a difference of 0.58 percentage points.

The Chi-square test confirms a significant difference across hours ($\chi^2(23) = 403.33$, $p < 0.001$, Cramér’s $V = 0.0267$), and pairwise tests with Bonferroni correction show that hour 16 differs significantly from hour 13 ($p < 0.001$), while hours 14–20 form a non-significant cluster. Although statistically significant, Cramér’s $V = 0.0267$ indicates a negligible practical effect, suggesting that the difference in conversion rates across hours is minimal in absolute terms.

Conversion Rate by Hour (ad group, Top 10)			
Hour	n	Conversion Rate	Std. Residual (TRUE)
13	45,485	2.51%	−0.64
14	43,779	2.86%	+3.96
15	42,855	2.98%	+5.57
16	35,963	3.09%	+6.34
17	33,605	2.85%	+3.43
18	31,052	2.75%	+2.12
19	29,169	2.68%	+1.35
20	27,846	3.03%	+4.94
21	28,895	2.92%	+3.86
22	25,515	2.65%	+0.91

Table 2: Conversion Rate and Standardized Residuals by Hour, Top 10 (ad group)

6 Bayesian Results

A randomization check using the Gamma-Poisson model confirms that the two groups exhibit comparable ad exposure, with $P(ad > psa) = 97.19\%$ and a 95% HDI of $[-0.0013, 0.1284]$ that includes zero, indicating no meaningful difference in *total.ads* between groups. Figure 8 shows the posterior distributions of the average ad exposure rate for both groups.

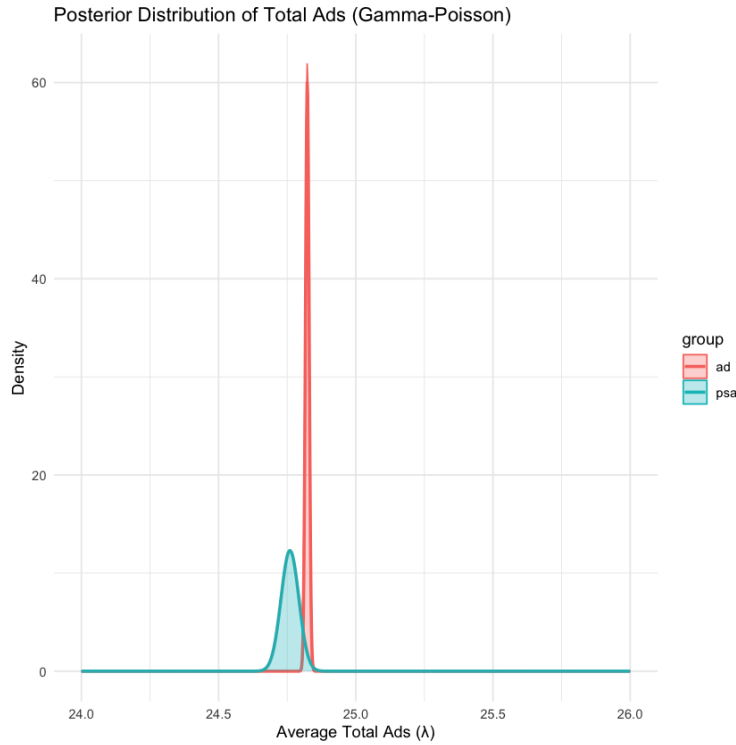


Figure 8: Posterior Distribution of Total Ads Rate (Gamma-Poisson)

6.1 Primary Analysis (converted: Beta-Binomial)

The Beta-Binomial model yields a posterior probability of $P(ad > psa) = 1.0000$, with a 95% HDI of $[0.0059, 0.0094]$ that excludes zero, confirming that the *ad* group achieves a higher conversion rate with near certainty. The posterior means are 2.55% for *ad* and 1.79% for *psa*. Figure 9 shows the posterior distributions of the conversion rate for both groups, with the *ad* distribution shifted clearly to the right.

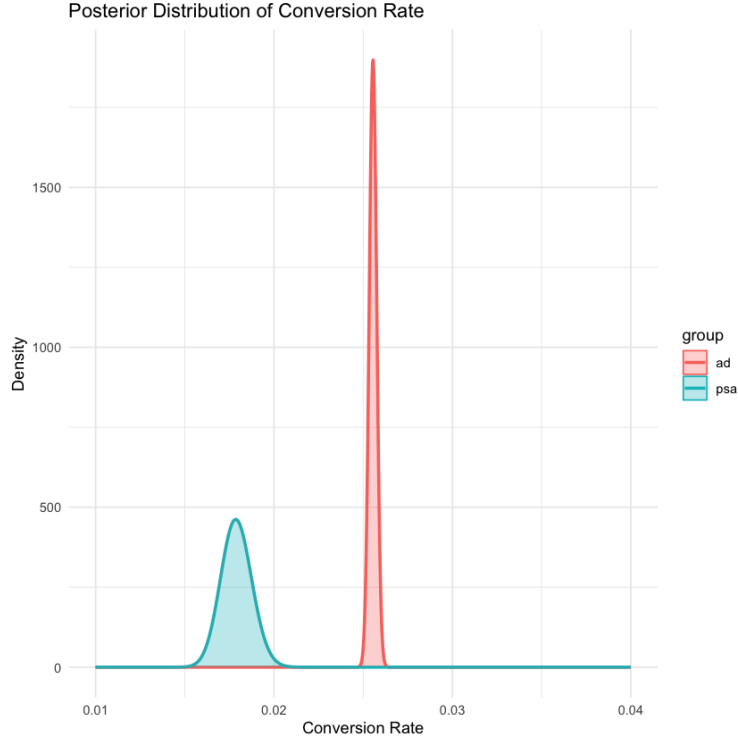


Figure 9: Posterior Distribution of Conversion Rate (Beta-Binomial)

6.2 Subgroup Analysis (ad group only)

Table 3 presents the posterior means and 95% HDI of conversion rates by day within the *ad* group. Monday achieves the highest posterior mean (3.33%), while Saturday records the lowest (2.13%). The non-overlapping HDIs between Monday and the lower-performing days confirm that the differences are credible rather than due to sampling variability.

Pairwise posterior comparisons show that Monday outperforms all other days with near certainty, including Tuesday ($P = 0.9993$, 95% HDI: [0.0011, 0.0045]), while Thursday, Friday, and Saturday show no meaningful difference among themselves. Figure 10 shows the posterior distributions for the three most contrasting days.

Posterior Conversion Rate by Day (ad group)			
Day	Posterior Mean	HDI Lower	HDI Upper
Monday	3.33%	3.20%	3.45%
Tuesday	3.05%	2.92%	3.17%
Wednesday	2.54%	2.43%	2.65%
Thursday	2.17%	2.07%	2.27%
Friday	2.25%	2.15%	2.34%
Saturday	2.13%	2.03%	2.23%
Sunday	2.46%	2.36%	2.57%

Table 3: Posterior Conversion Rate by Day (ad group)

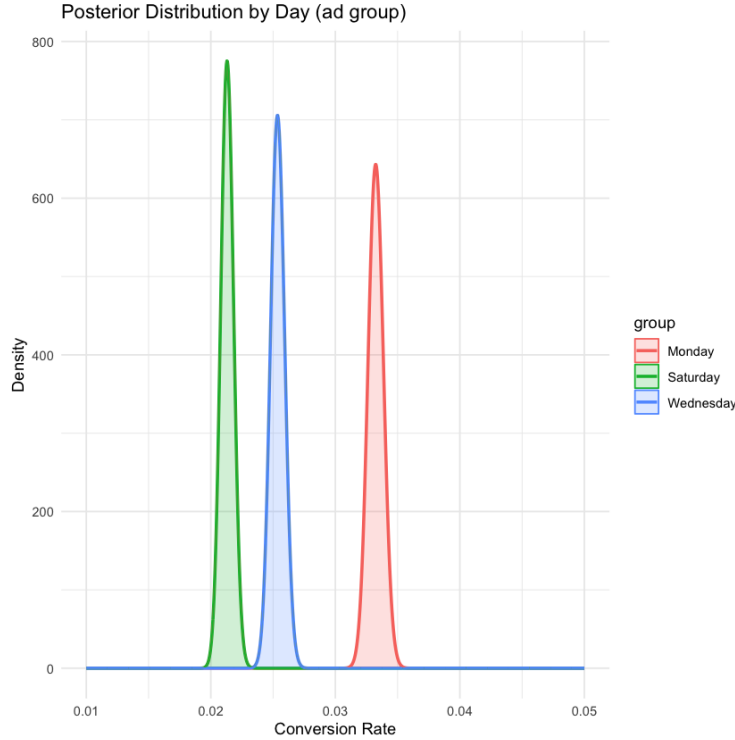


Figure 10: Posterior Distribution of Conversion Rate by Day (Monday, Wednesday, Saturday)

Table 4 presents the posterior means and 95% HDI of conversion rates by hour within the *ad* group, focusing on the top 10 performing hours (13:00–22:00). Hour 16 achieves the highest posterior mean (3.09%), while hour 13 records the lowest (2.51%).

Pairwise posterior comparisons show that hour 16 outperforms hour 13 with near certainty ($P = 1.0000$, 95% HDI: [0.0036, 0.0082]), while hour 16 shows no meaningful difference from

hours 15, 20, and 21 (HDI includes zero), suggesting that the afternoon peak (15:00–21:00) performs comparably.

Posterior Conversion Rate by Hour (ad group, Top 10)			
Hour	Posterior Mean	HDI Lower	HDI Upper
13	2.51%	2.37%	2.65%
14	2.86%	2.70%	3.02%
15	2.99%	2.83%	3.15%
16	3.09%	2.91%	3.27%
17	2.86%	2.68%	3.04%
18	2.75%	2.57%	2.93%
19	2.68%	2.50%	2.87%
20	3.03%	2.83%	3.23%
21	2.92%	2.72%	3.11%
22	2.65%	2.45%	2.85%

Table 4: Posterior Conversion Rate by Hour, Top 10 (ad group)

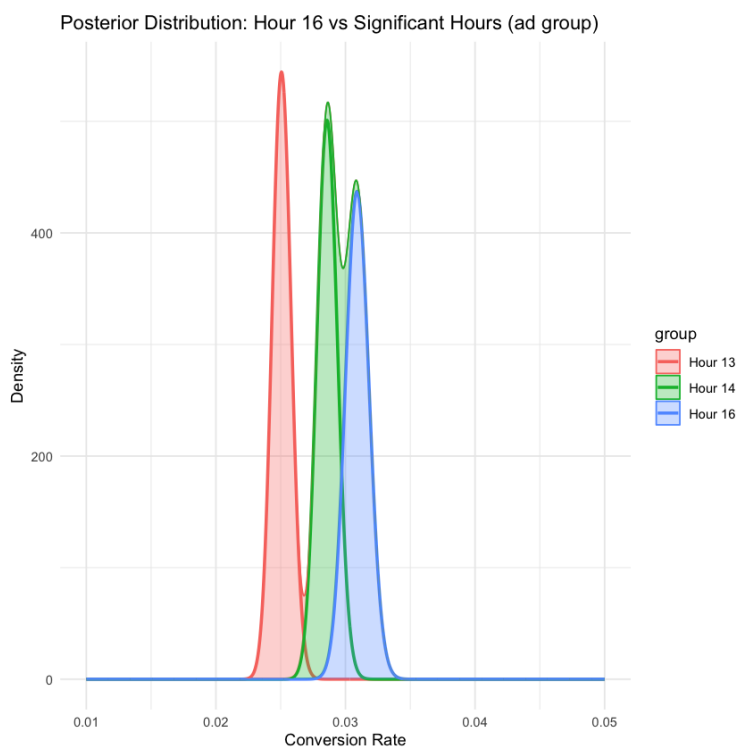


Figure 11: Posterior Distribution of Conversion Rate by Hour (Hour 13, 14, 16)

7 Overall Comparison

Both Frequentist and Bayesian approaches consistently confirm that advertisement exposure significantly increases conversion rate. The Frequentist two-sample proportion test yields $p < 0.001$, while the Bayesian Beta-Binomial model estimates $P(ad > psa) = 1.0000$ with a 95% HDI of $[0.0059, 0.0094]$ excluding zero. However, both approaches agree that the practical effect is negligible (Cohen’s $h = 0.053$).

For subgroup analysis, both approaches identify Monday and the afternoon hours (14:00–20:00) as the highest-performing conditions, with Hour 16 achieving the peak conversion rate. All effect sizes remain negligible, consistent with the large sample size inflating statistical significance.

8 Discussion & Conclusion

8.1 Key Findings

Both Frequentist and Bayesian analyses consistently confirm that advertisement exposure has a statistically significant positive effect on conversion rate. The *ad* group achieves a conversion rate of 2.55%, compared to 1.79% for the *psa* group, with near-certain posterior probability ($P = 1.0000$). Despite statistical significance, all effect sizes remain negligible, reflecting a common pattern in large-sample A/B tests where even trivial differences reach significance.

8.2 Business Recommendations

Based on the analysis, advertisement exposure is recommended to be prioritized on **Monday** and during **afternoon hours (14:00–20:00)**, particularly at **hour 16**, where conversion rates are consistently highest.

Users with high ad exposure ($total.ads > 96$) exhibit substantially higher conversion rates (17.18%), suggesting that sustained advertisement exposure has a cumulative effect on purchase likelihood. Advertisers may therefore benefit from increasing ad frequency for engaged users rather than broadening reach to new audiences.

8.3 Limitations

Dataset Limitations

The dataset does not include contextual variables such as user demographics, device type, or advertisement content, which may confound the relationship between ad exposure and conversion. Additionally, it is unclear whether *total.ads* represents unique advertisement impressions or repeated exposures to the same advertisement, limiting the interpretability of the cumulative exposure effect.

Evaluation Metrics

The analysis relies solely on *converted* as the primary outcome metric. While conversion captures whether a purchase occurred, it does not reflect revenue per user or customer lifetime value, which would provide a more comprehensive evaluation of advertisement effectiveness. Furthermore, all effect sizes remain negligible despite statistical significance, suggesting that the practical impact of the findings should be interpreted with caution.

References

[1]